

PID Simulation Construction Report

Version: 2.1

Date: 2026-08-04

Scope: Document the construction, assumptions, and validation of `uavx-python/src/tests/test_pid_sim.py` — the calibrated PID simulator used for the FW/MR tuning studies. **Supersedes v1** `FW_SIMULATION_REPORT.md` (2026-07-26, “Aerodynamic Model v1”, `qbar×S×C_ctrl` physics without the 70%-deflection authority anchor). This is the authoritative construction reference.

v2.0 → v2.1 (2026-08-04): Added `is_roll_structurally_limited()` — aileron-less RudderElevatorAF frames have zero roll authority in the coupled model (roll routed to rudder, bank-and-yank cancelled by the yaw-angle hold). Roll step/gust criteria are non-scored for these frames, mirroring the existing yaw gate for rudderless elevons. See §7.

1. Purpose

Simulate each airframe’s closed-loop step response (attitude, disturbance rejection, altitude hold, navigation) so PID gains can be tuned against physically meaningful authority before flight. The sim models the **full control path**: PID loops → FC mixer (DoServos/DoMotors) → aerodynamic/motor physics.

2. Architecture

```
run_tests_for_af(af_filename)
    get_params_for_af() → params, AirframeType, FW model idx, is_fw
    FW (descriptor present): simulate_axis_coupled() - 3-axis coupled aero
    FW/MR (legacy):         simulate_axis()         - single-axis
    Disturbance rejection:   simulate_rate_disturbance()
    Altitude hold:           simulate_alt_hold_fw() / simulate_alt_hold_mr()
    Navigation:              simulate_nav_fw() / simulate_nav_mr()
    Free-flight turbulence:  simulate_freeflight() (FW only)
```

Active engine lives at `uavx-python/src/tests/test_pid_sim.py`. Copies under `windows/`, `linux/`, `macos/` are stale snapshots — always run the `uavx-python` copy.

3. Parameter/Physics Sources

- **Params:** parsed from the `.af` file (`airframes.airframes.parse_af_file`) → `{ParamIndex: raw_float}` → `load_af_params()` names.
 - **FW physicals:** `.af` `PHYS_*` metadata merged over `FW_AIRFRAMES` defaults by `get_fw_descriptor()` (`PHYS_TO_DESCRIPTOR`). The `.af` is authoritative.
 - **MR physicals:** `PHYS_AUW_G`, `PHYS_ARM_MM`, `PHYS_MOTOR_COUNT`, `PHYS_MOTOR_THRUST_G` → hover/max thrust, arm length, inertia per frame (fallback `EM_*` `emu.c` defaults).
 - **Enums:** `protocol_enums.ParamIndex`, `AirframeType`, `AF_CATEGORY`, `FW_MODEL_IDX`, `AF_FILES`.
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4. FW Physics (simulate_axis_coupled, run_physics_fw)

Control torques

$L = C_{l_ail} \cdot \dot{\alpha}$ $C_{l_ail} = \bar{q} \cdot S \cdot b \cdot CL_{D_AIL}$
 $M = C_{m_ele} \cdot \dot{\epsilon}$ $C_{m_ele} = \bar{q} \cdot S \cdot c \cdot CM_{D_ELE}$
 $N = C_{n_rud} \cdot \dot{r}$ $C_{n_rud} = \bar{q} \cdot S \cdot b \cdot CN_{D_RUD}$
 $N_{elevon} = \text{adverse_yaw} \cdot C_{l_ail} \cdot \dot{\epsilon}_{elevon}$ (ElevonAF drag-differential)

Damping

quadratic: $C_{d_axis} \cdot r \cdot |r|$ (PHYS*_DAMP)
linear: $C_{lin_axis} \cdot |r|$ (*_damp_lin)

Stability & coupling

- Pitch/yaw static stability: $\pm \text{stab_coeff} \cdot \bar{q} \cdot S \cdot \text{ref} \cdot \text{angle}$
- Adverse yaw: $N_{\text{adverse}} = \text{adverse_yaw} \cdot L_{\text{ctrl}}$
- Dihedral: $L_{\text{dihedral}} = \text{dihedral_coeff} \cdot r_{\text{yaw}}$
- Servo lag: first-order servo_tau on all surface channels

Rate ceiling

_compute_fw_max_rates() solves the quadratic damping balance at full deflection ($\approx 1.195 \times \text{MAX_*_RATE}$). The sim clamps rates to this ceiling; **MAX*_RATE is the 70%-deflection steady-state rate, not the clamp.**

Mixing (FC DoServos/DoMotors)

Efforts R/I/P/YI route per AF type — see §6 of FW_Control_Authority_Calibration.md. Elevon, Rudder/Elevator, Delta/Aileron/Spoileron paths all modeled; feedforwards FW_ROLL_PITCH_FF, FW_AILERON_RUDDER_MIX applied.

5. MR Physics (run_physics_mr, simulate_axis)

- Torque from motor differential thrust: $\tau = kT \cdot \text{effort}$, $kT = \text{max_thrust} \cdot 0.25 \cdot \text{arm}$.
 - Axis damping: Roll 0.015, Pitch 0.03, Yaw 0.05 (quadratic, sign-opposed).
 - Motor lag: first-order EM_MOTOR_TAU (0.10 s).
 - Inertia from arm/mass per frame (shape factor by motor count: 4→0.55, 6→0.40, 8→0.30).
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6. Outer Loops

Altitude Hold

- **MR** (simulate_alt_hold_mr): position PI → desired ROC → velocity PI → throttle comp. Fast tilt response.
- **FW** (simulate_alt_hold_fw): position PI (ALT_POS_KP/KI) → ROC setpoint → velocity PI (ALT_ROC_KP, UNUSED_ALT_VEL_KI) → climb with climb_tau = 2.0 s aero lag. Airspeed-bleed coupling.

Navigation

- **MR** (`simulate_nav_mr`): cross-track error → NAV_POS_KP/KI → velocity → bank (NAV_VEL_KP → $g \cdot \tan(\text{bank})$) — fast (~ 1 s).
 - **FW** (`simulate_nav_fw`): cross-track → NAV_POS_KP/KI → bank-angle command → coordinated turn $\text{heading_rate} = g \cdot \tan(\text{bank})/V$ — slower ($\text{bank_tau} = 1.5$ s).
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7. Criteria

Test	Roll	Pitch	Yaw
MR attitude (CRITERIA)	rise ≤ 1.0 s, OS $\leq 10\%$, settle ≤ 2.5 s, err $\leq 1^\circ$	same	rise ≤ 3 s, OS $\leq 15\%$, settle ≤ 5 s, err $\leq 2^\circ$
FW attitude (FW_CRITERIA)	rise ≤ 4 s, OS $\leq 25\%$, settle ≤ 8 s, err $\leq 5^\circ$	rise ≤ 5 s, OS $\leq 30\%$, settle ≤ 12 s, err $\leq 15^\circ$	rise ≤ 5 s, OS $\leq 25\%$, settle ≤ 10 s, err $\leq 5^\circ$
Disturbance (DIST_CRITERIA)	peak $\leq 15^\circ/\text{s}$, settle ≤ 1.5 s, IE $\leq 10^\circ$	peak $\leq 12^\circ/\text{s}$, IE $\leq 8^\circ$	peak $\leq 12^\circ/\text{s}$, IE $\leq 10^\circ$
AH (MR/FW)	AH_CRITERIA_MR rise ≤ 2 s OS $\leq 20\%$ settle ≤ 5 s err $\leq 1^\circ$	AH_CRITERIA_FW rise ≤ 6 s OS $\leq 15\%$ settle ≤ 8 s err $\leq 2^\circ$ (climb-tau floor ~ 4.4 s)	—
Nav (MR/FW)	rise ≤ 2 s/3s, OS $\leq 20\%/15\%$, settle ≤ 5 s, err $\leq 1^\circ/2^\circ$	—	—

Structural exclusions: rudderless elevons (`is_yaw_structurally_limited`) — yaw step/gust non-scored; aileron-less RudderElevatorAF frames (`is_roll_structurally_limited`) — roll step/gust non-scored.

8. Free-Flight (FW)

Open-loop 8s turbulence response with zero surface deflection — measures **intrinsic** stability (dihedral, static margin, damping) without PID. Reports roll/pitch/yaw peak excursions. A large roll divergence (e.g. Shadow 1286°) is a **structural** characteristic of low-dihedral flying wings, not a tuning failure.

9. Known Gaps / Caveats

1. **Rudder-only roll** (`is_roll_structurally_limited`, v2.1): `CL_D_AIL = 0` → zero aileron authority → roll ceiling 0. Bank-and-yank roll authority under-modeled (rudder produces yaw → dihedral couples to roll), so roll is **non-scored** for these frames rather than reported as an unreachable FAIL.
2. **Elevon yaw:** drag-differential authority is approximate (~ 1 – 3° hold). No differential-motor yaw (all FW frames single-motor; `RudderMotorFF` path exists in FC but no multi-motor FW frame in the set).

3. **Airspeed coupling:** AH/Nav FW use fixed cruise speed; no throttle-induced pitch coupling beyond `climb_tau`.
 4. **DT/EM mismatch:** windows/, linux/, macos/ sim copies are stale — the authoritative file is `uavx-python/src/tests/test_pid_sim.py`.
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10. Files Modified (this calibration)

- `uavx-python/src/tests/test_pid_sim.py` — `get_fw_descriptor`, `load_af_physicals`, `quadratic_compute_fw_max_rates`, `PHYS_TO_DESCRIPTOR`, `C_d_roll` from `PHYS_ROLL_DAMP` (3 call sites), `elevon yaw mixing`, `criteria relax`, **v2.1** `is_roll_structurally_limited` + `Roll gating` in `run_tests_for_af`.
- 12 FW `.af` files — `PHYS_*` `aero block` + `RATE_KP = 0.7/MAX*_RATE`.
- Reports: `wiki/FW_Control_Authority_Calibration.md` (this series), `wiki/tuning_study.md`.